

DSDL Midterm Cheat Sheet

B13902022 Yu-Chi Lai

Number Representation

Let n be word length (bits).

- **Sign-magnitude (SM):** 1 sign bit + $(n - 1)$ magnitude bits.
- **1's comp.** $\bar{N} = (2^n - 1) - N$ (flip all bits).
- **2's comp.** $N^* = 2^n - N = \bar{N} + 1$.
- **Sign extension:** copy sign bit when increasing word length.

For $n = 4$: SM/1's comp range $[-7, +7]$ with ± 0 ; 2's comp range $[-8, +7]$.

Complement Addition

2's comp:

- Add as unsigned, discard carry out of MSB.
- Overflow iff same-sign inputs give opposite-sign output.
- Equivalent test: $V = C_{in \rightarrow MSB} \oplus C_{out MSB}$.

Reason: terms contain an extra 2^n that is dropped modulo 2^n .

1's comp:

- Add as unsigned, then add end-around carry to LSB.
- Same overflow rule as above.
- Has two zeros: $+0 = 00 \dots 0$, $-0 = 11 \dots 1$.

Boolean Algebra

Uniting: $XY + XY' = X$, $(X + Y)(X + Y') = X$

Absorption: $X + XY = X$, $X(X + Y) = X$

Elimination: $X + X'Y = X + Y$, $X(X' + Y) = XY$

Consensus: $XY + X'Z + YZ = XY + X'Z$

$(X + Y)(X' + Z)(Y + Z) = (X + Y)(X' + Z)$

Distributive: $X(Y + Z) = XY + XZ$, $(X + Y)(X + Z) = X + YZ$

De Morgan: $(XY)' = X' + Y'$, $(X + Y)' = X'Y'$

Useful: $(X + Y)(X' + Z) = XZ + X'Y$

Duality: swap $0 \leftrightarrow 1$, AND $\leftrightarrow$ OR; keep variables/complements.

Canonical Forms & Conversion

For n variables, define universe $U = \{0, \dots, 2^n - 1\}$.

If $F = \sum m(S)$, then

$$F = \prod M(U \setminus S), \quad F' = \sum m(U \setminus S) = \prod M(S).$$

Let $A, B \subseteq U$:

- $\sum m(A) \cdot \sum m(B) = \sum m(A \cap B)$
 - $\prod M(A) \cdot \prod M(B) = \prod M(A \cup B)$
 - $\sum m(A) + \sum m(B) = \sum m(A \cup B)$
 - $\prod M(A) + \prod M(B) = \prod M(A \cap B)$
- Example: $F_1 = \sum m(0, 2, 3, 5, 9, 11)$, $F_2 = \sum m(0, 3, 9, 11, 13, 14)$

$$F_1 F_2 = \sum m(0, 3, 9, 11).$$

Convert expression to canonical form:

- Use truth table directly.
- SOP $\rightarrow$ minterms: insert missing vars with $(X + X') = 1$ and expand.
- POS $\rightarrow$ maxterms: insert missing vars with $XX' = 0$ and factor.

QM Method

1. List all minterms.
2. Include don't-care terms for grouping only.
3. Combine adjacent terms; mark covered terms.
4. Repeat until no further combining.
5. Build prime implicant (PI) chart.
6. Keep essentials first.
7. Cover remaining minterms with minimum PIs.

K-map Simplification (Recap)

Minimum SOP = minimum cover = smallest PI set that covers all 1-cells (first minimize PI count, then literal count).

1. Mark 1's (and optional don't-cares).
2. Form largest possible power-of-2 groups (wrap-around allowed) to find PIs.
3. Pick all essential PIs.
4. Cover uncovered 1's with as few extra groups as possible (minimum cover).

Multilevel Gate Circuits

SOP form chain:

- AND-OR $\leftrightarrow$ NAND-NAND
- NAND-NAND $\leftrightarrow$ OR-NAND
- OR-NAND $\leftrightarrow$ NOR-OR
- NOR-OR $\leftrightarrow$ AND-OR

POS form chain: OR-AND $\leftrightarrow$ NOR-NOR $\leftrightarrow$ AND-NOR $\leftrightarrow$ NAND-AND $\leftrightarrow$ OR-AND.

Cost bounds:

- gates $\leq$ terms + 1,
- gate inputs $\leq$ literals + terms.
- Equality can fail when a term degenerates to one variable.

Hazards

- **Static-1:** same check as 2-level logic.
- **Static-0:** pattern $XX'\alpha$ in SOP, adjacent 0-cells differ in X with $\alpha = 1$.
- **Dynamic:** for adjacent input states with opposite outputs, change in X reaches ≥ 3 paths.
- Practical fix: add redundant consensus term to cover adjacent cells.

Hazard-free AND-OR: use SOP where every adjacent 1-pair is covered by at least one implicant. OR-AND uses the dual rule.